SymPy Bug Card

Bug 17: telescoping sum ignores undefined initial terms

Task. Evaluate the infinite sum

$$\sum_{k=0}^{\infty} \frac{1}{k(k-1)}.$$

SymPy's answer.

$$\mathrm{Sum}\left(\frac{1}{k(k-1)}, (k, 0, \infty)\right).\mathrm{doit}() = -1.$$

Correct answer.

$$\sum_{k=0}^{\infty} \frac{1}{k(k-1)} \text{ is undefined.}$$

Explanation. The terms at $k = 0$ and $k = 1$ have denominator zero, so although $\frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$ telescopes on ranges where the terms are defined, the requested range begins with two undefined terms. The well-defined tail from $k = 2$ sums to 1, so the returned value -1 is not a valid value for the tail either.

Likely root cause. The diagnosis points to the telescoping path in `sympy/concrete/summations.py`: after partial fractions, SymPy applies the endpoint telescoping formula without checking for poles in the original summation range. This is a new instance of a known failure mode.

Artifact (GitHub).
`bug-report/bug-017-telescoping-sum-ignores-undefined-initial-terms/`